BROOKBANKS

RM1 Primary Street Woodlands, NEV

SURFACE WATER DRAINAGE STRATEGY

Document Control Sheet

Document Title	Drainage Strategy
Document Ref	1335 DS 101 Rv1
Project Name	Woodlands
Project Number	1335
Client	Vistry/Taylor Wimpey

Document Status

Rev	Issue Status	Prepared / Date	Checked / Date	Approved / Date
0	Client comment	SM/CM 06.08.24	NO 06.08.24	
1	LLFA comment	SM 24.12.24	NO 18.02.25	

Issue Record

Name / Date & Revision	12/08/24					
DLA	-					

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Executive Summary

1.1 Brookbanks is appointed by Taylor Wimpey and Vistry to provide support for the strategic drainage design at the Woodlands Site in Swindon. This Reserved Matters application (following planning application S/OUT/13/1555) includes details of the access, appearance, layout, scale and landscaping in relation to the primary street through South Marston including discharge of conditions 11 (Landscape Ecology and Arboricultural Management Plan), 13 (Tree Protection), 17 (Bats), 31 (Movement Phasing Plan), 36 (Construction Method Statement), 37 (Construction Environmental Management Plan), 42 (Surface Water Drainage), 53 (Slab Levels), 54 (Materials) and 56 (Street Lighting and Street Furniture).

1.2 The site already benefits from an Outline Planning Consent (ref. S/OUT/13/1555) which covers the principles of the development and associated infrastructure within the site boundary along with the approved foul drainage strategy and surface water drainage strategy Condition 41 Report ref SWDR Rev 5. This report provides supporting details to demonstrate that the development remains acceptable from a flood risk and drainage viewpoint and accords with the previously approved Flood Risk Assessment following further detailed design and includes the following.

- Detailed Drainage plans, including proposed site levels, impermeable areas and exceedance flow routes.
- Details of the drainage features, including sections
- Detailed Drainage calculations to support the design for all rainfall events up to and including the 1 in 100-year rainfall event (including an allowance for climate change)
- Details of Adoption and whole life Maintenance requirements in line with current best practice
- Details of Pollution/Water Quality.

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2 Site Wide Drainage Strategy

2.1 The approved strategic surface water drainage scheme is set out in Brookbanks Technical Note SWDR rev 5 dated January 2024 the report was prepared to discharge Condition 41 with respect to the outline planning application (ref S/OUT/13/1555)

2.2 The surface water drainage scheme is based on the principles set out in the Flood Risk Assessment (FRA) (ref. 1335/FRA/03). The FRA was submitted and approved as part of the outline planning application and the Surface Water Management Concept Plan 1335-DR-05 C, the agreed starting point for the strategy insofar as dealing with existing overland flows and proposed SUDS.

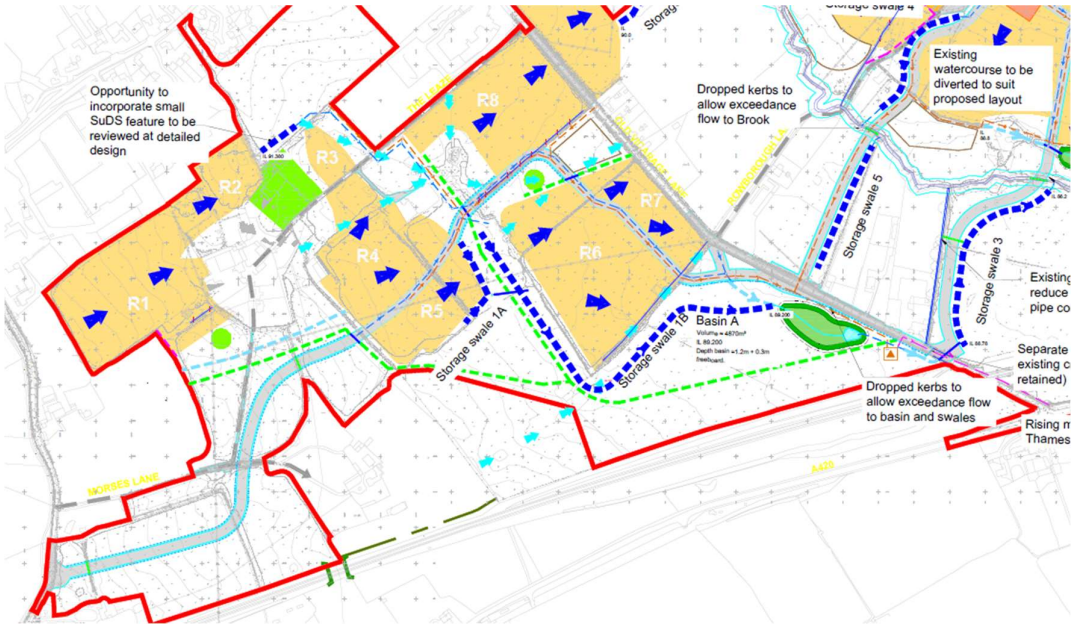

Figure 2-1: Extract from proposed drainage strategy 1335-SK-114, South Marston.

2.3 In summary the site wide strategy set out the following principles

- strategy for storm drainage at the site has been developed to meet both national and local policy. The proposals employ means of drainage to comply with NPPF guidance, together with SFRA and other national and local guidance.
- The drainage proposals will manage storm water by way of a SUDS management train and ensure peak discharges from the developed land provide betterment from the appraised baseline rates.
- The proposals will also provide to maintain the quality of water discharged from the development during the construction and operational stages.
- The storm water management proposals will improve the amenity and biodiversity of the site.
- The storm water system has been designed so that it is suitable for adoption by TW or NAV for future maintenance. Future maintenance of the non- adoptable SuDS elements of the system will be undertaken by a private management company.
- Implementation of the storm water management system for each development parcel will be undertaken prior to first occupation of dwellings.

3 Proposed Surface Water Drainage Strategy

Development Description

Reserved Matters application (following planning application S/OUT/13/1555) for details of the access, appearance, layout, scale and landscaping in relation to the primary street through South Marston including discharge of conditions 11 (Landscape Ecology and Arboricultural Management Plan), 13 (Tree Protection), 17 (Bats), 31 (Movement Phasing Plan), 36 (Construction Method Statement), 37 (Construction Environmental Management Plan), 42 (Surface Water Drainage), 53 (Slab Levels), 54 (Materials) and 56 (Street Lighting and Street Furniture).

3.1 The surface water drainage strategy has been developed in accordance with the Site Wide parameters set out in Section 2.

3.2 A full set of detailed drawings and calculations have been developed to support the RMA and are attached as Appendix B and C

Existing Drainage

3.3 Existing ditchcourses and overland flow routes will be maintained as set out in the site wide strategy.

3.4 The overland flow routes will be formed as scrapes in the ground to route flows to the existing outfall points along the western edge of the site. Where the existing ditchcourses run under proposed highway we have designed piped culvert crossings to accommodate anticipated flow and ecological requirement.

3.5 Detailed drawings are enclosed in Appendix D

Proposed Drainage

3.6 The proposed drainage for RM1 will consist of the following.

- Gullies and sewers in the residential frontage areas.
- Source control in the form of over the edge swales, the swales will be under drained by carrier pipes.
- Secondary swales located in green areas providing storage, conveyance and treatment.
- Basin A located in the southern corner to attenuate for up to the 1 in 100-year storm event plus climate change.
- Outfall to the existing ditch network as an interim measure subject to a discharge consent with the LLFA.

3.7 The proposed highway corridor has been designed to ensure that all drainage is by gravity, according to adoptable standards. Design levels are indicated in the proposed contour plans included in **Appendix B.**

3.8 The proposed catchments have been updated from the site wide strategy to reflect the fully detailed highway corridor. The impermeable areas for the residential catchments are as set out in the Site Wide Strategy, further details for the parcel drainage strategy will be provided with each supporting RMA.

3.9 Detailed catchment plans are included. Gully spacing has been calculated based on Swindon Borough Council guidance. Residential catchment assumptions remain as set out in the site wide strategy and summarised below.

Basin	Catchment Area	Impermeable Area + UC (ha)	Proposed 100 Year Run-off (l/s)	Detention Volume for 1 in 100 Year (+40%cc)
A	R1, R2, R3, R4, R5, R6, R7 & Primary Infrastructure	8.47	38.9	4,868

3.10 A full set of Flow software calculations are included in Appendix C. The calculations show that flow from the RM site is 38.9 l/s for storms up to and including the 1 in 100-year storm event plus 40 % climate change.

3.11 In accordance with the site wide strategy the detention basin accords with the basic principles

- Design based on principles outlined in CIRIA Report 753 - The SuDS Manual.
- Permanently wetted areas.
- Ecological and landscape considerations
- Shallow and varied side slopes.
- Provide water treatment and capture of accidental spillages.
- Native emergent and fringing aquatic water purifying planting.
- Biodiversity and amenity opportunities maximised where possible.
- Access provided throughout for maintenance purposes.

- Outfalls modelled as surcharged as appropriate with water level/flood data.

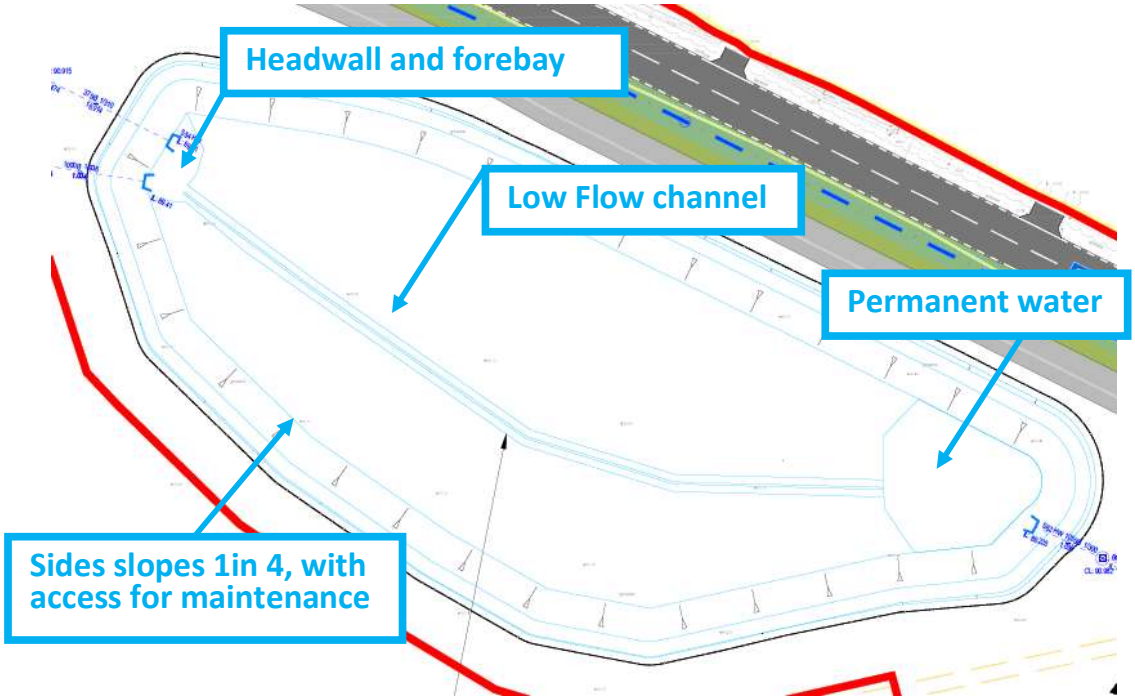

Figure 3-1: Proposed detention basin A

3.12 Full details of swales, basins, under drained swales, headwalls and flow controls are included as Appendix D.

3.13 A foul sewer will be sited within the adopted highway, outfall to the proposed foul pumping station to the southern corner of the site subject to a separate reserved matters application. The foul drainage is designed to adoptable standards in accordance with the DCG.

Exceedance Flows

3.14 The drainage calculations demonstrate the surface water system is designed to adoptable standards and will not flood in the 1 in 30-year event.

3.15 The likely exceedance routes have been plotted on the proposed layout and included in Appendix B, as per the approved drainage strategy dropped kerbs and land profiling will ensure flows are routed away from residential areas to the natural drainage routes.

Outfall

3.16 The proposed outfall will be to an existing ditchcourse. The contractor will clear and regrade to ensure connectivity to the existing South Marston Brook. As the Rowborough site is implemented, this outfall will be connected to storage swale 3 as per the approved strategy.

3.17 The likely exceedance routes have been plotted on the proposed layout and included in **Appendix B.** As per the approved drainage strategy dropped kerbs and land profiling will ensure flows are routed away from residential areas to the natural drainage routes.

4 Maintenance Schedule

4.1 The proposed drainage network will be adopted by an appointed NAV and subject to S104 technical approval.

4.2 Highway drainage will be managed by Swindon Borough Council following adoption under a S38 Agreement.

4.3 Unadopted SuDS features will be maintained by a management company or offered to the Parish Council for maintenance.

4.4 The developer and the appointed contractor will be responsible for all surface water management during construction and maintenance period. Temporary drainage works will be implemented in line with the CEMP.

4.5 Further detail of the proposed maintenance arrangements is set out below.

Gullies and Sewers

4.6 In places surface water run-off will be captured and conveyed via a conventional drainage system consisting of pipes, manholes and gullies.

4.7 Runs in the future adoptable system are connected using manholes to DCG whilst pipe runs serving the private plots are connected using inspection chambers.

4.8 Manholes, inspection chambers and rodding eyes are provided at the termination or change in direction or gradient of the pipes runs to allow for routine inspection and maintenance.

4.9 Gullies are to be trapped to minimise sediment entry into the system and prevent odours returning from the sewers.

4.10 During the construction works the Contractor will be responsible for the inspection and maintenance of any elements of the drainage.

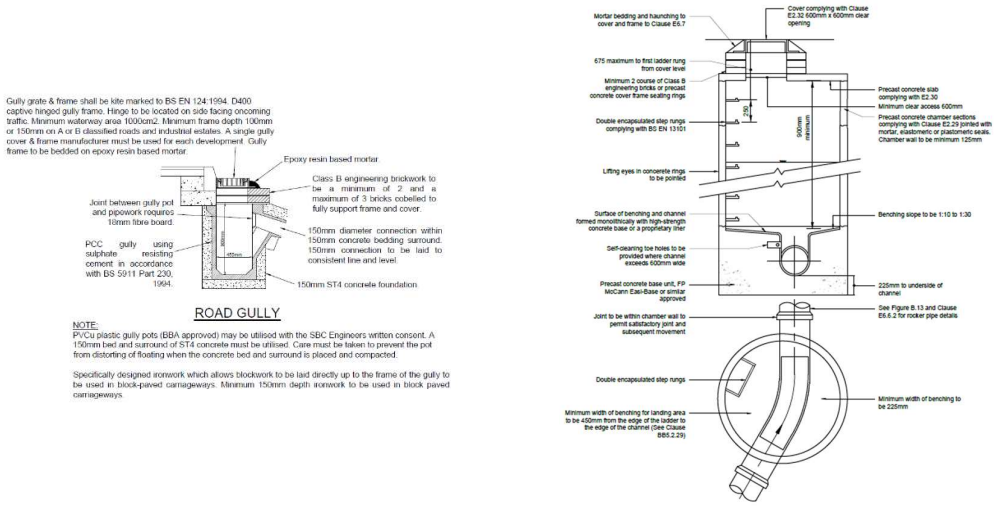

Figure 4-1: Typical Gully & Manhole

4.11 Inspection and maintenance to be carried out as detailed in the table below.

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Inspect and identify any areas that are not operating as required.	Monthly (for the first 3 months) then annually.
	Remove debris from gratings, sumps / silt traps	Annually
Occasional Maintenance	Clear blockages	As required
Remedial Actions	Sumps / silt traps found to be non-functioning or blocked on annual inspection to be noted and cleared at 6-month intervals. Recurring blockages to be investigated	Annually (or 6 months if blocked annually).
	Cleanings / jetting of pipes, channel drains, chambers and manholes.	As required

Table 4-1: Conventional Drainage Maintenance Schedule

Basins and outfalls.

4.12 The inlet and outlet headwalls for the basins are to be pre-cast concrete headwalls with suitable protection measures preventing both unwanted access to the pipes and falling hazards from the top of the headwall.

4.13 The basin is designed to be1.5m deep with side slopes varying from 1 in 5 to 1 in 3 to ensure accessibility for maintenance. The maximum water level in extreme events is 1.2m. The basin will be in open landscaped areas and will not be fenced off, information boards and safety features will be included to inform public.

4.14 The basin will contain a sunken area of permanent water prior to the outlet headwall. This will serve as a sediment trap and will contain planting to provide treatment to the water.

Figure 4-2: Typical Detention Basin - Source: CIRIA C753 "The SuDS Manual".

4.15 Inspection and maintenance to be carried out as detailed in the table below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Remove litter and debris	Monthly or as required
	Cut grass – for spillways and access routes	Monthly or as required
	Cut grass – meadow grass in and around basin	Half yearly (spring before nesting season, and autumn)
	Manage other vegetation and remove nuisance plants	Monthly at the start and then as required
	Inspect inlets, outlets and overflows for blockages, and clear if required	Monthly
	Inspect banksides, structures, pipework etc for signs of physical damage	Monthly
	Inspect inlets and facility surface for silt accumulation, establish appropriate silt removal frequencies	Monthly (for first year), then annually or as required.
	Check penstocks and other mechanical devices	Annually

	Tidy all dead growth before start of growing season	Annually
	Remove sediment from inlets, outlets and permanently wet area	3 months (inspection) 6 months (cleaning), or as required
	Manage wetland plants in outlet pool	Annually
Occasional Maintenance	Reseed areas of poor vegetation growth	As required
	Prune and trim trees and remove cuttings	Every 2 years or as required
	Remove sediment from inlets, outlets, permanently wet area, and main basin when required	Every 5 years, or as required.
Remedial Actions	Repair erosion or other damage by re-turfing or reseeding	As required
	Repair inlets & outlets	
	Relevel uneven surfaces and reinstate design levels	

Table 4-2: Basin Maintenance Schedule

Swales and watercourse

4.16 The swales will be predominantly dry with intermittent flows during rainfall events.

Figure 4-3: Typical Swale - Source: CIRIA C753 "The SuDS Manual".

4.17 Inspection and maintenance to be carried out as detailed in the table below:

Maintenance Schedule	Required Action	Typical Frequency

Regular Maintenance	Remove litter and debris	Monthly or as required
	Cut grass – to retain grass height within specified design range	Monthly (during growing season) or as required
	Manage other vegetation and remove nuisance plants	Monthly at the start, and then as required.
	Inspect inlets, outlets and overflows for blockages, and clear if required	Monthly
	Inspect vegetation coverage	Monthly for 6 months, quarterly for 2 years, then half yearly
	Inspect inlets and facility surface for silt accumulation, establish appropriate silt removal frequencies	Half Yearly
Occasional Maintenance	Reseed areas of poor vegetation growth, alter plant types to better suit conditions, if required	As required or if bare soil is exposed over 10% or more of the swale treatment area
Remedial Actions	Repair erosion or other damage by re-turfing or reseeding	As required
	Relevel uneven surfaces and reinstate design levels	
	Remove build-up of sediment on upstream gravel trench, low spreader or at top of filter strip	
	Remove and dispose of oils or petrol residues using safe standard practices	

Table 4-3: Swale Maintenance Schedule

Culverts

4.18 Existing ditchcourses will be culverted. These culverts will serve to allow the conveyance of the water along the watercourse underneath the primary road through the site and smaller footpath crossings.

4.19 The culverts will be formed from precast concrete headwalls connected with a concrete pipe.

4.20 Methods for the removal of debris from the culvert and screens are set out in ***CIRIA C786 Culvert, Screen and Outfall Manual***. In general, manual removal and cleaning or methods involving entry into confined spaces are to be avoided, with a preference for mechanical cleaning and removal. Suitable methods for removal to be determined by the maintaining party depending on the specific requirements at the time.

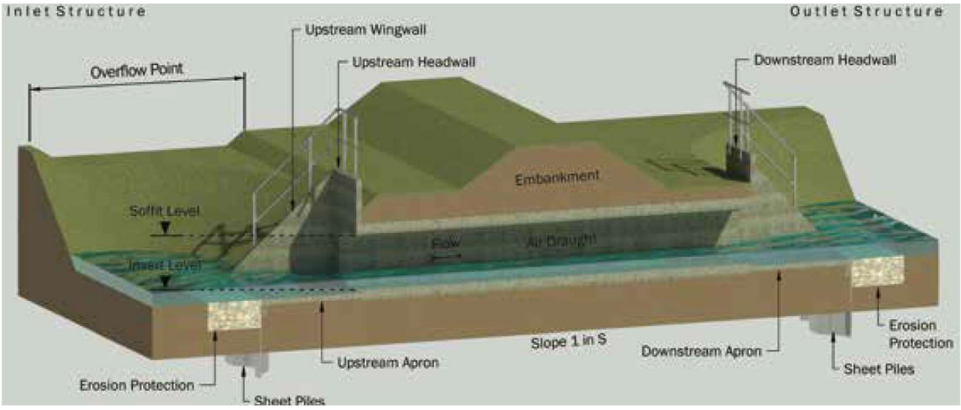

Figure 4-4: Typical Culvert - Source: CIRIA C786 Culvert, Screen and Outfall Manual

4.21 Inspection and maintenance to be carried out as detailed in the table below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Inspect and identify any blockages on the screens or within the pipe.	As required
Occasional Maintenance	Removal of debris from screens	As required
Remedial Actions	High pressure jetting	As required

Table 4-4: Culvert Maintenance Schedule

Flow Control Devices

4.25 Flow control devices are intended to throttle the discharge rate from the site, often for the purpose of replicating or improving upon the natural flow rate the land would have when undeveloped / improving upon the existing discharge rate for developed land.

4.26 The flow controls used on site will be vortex flow controls. These are designed to provide predictable flow rates for certain heads of water and, due to their non-linear flow rate, operate well in both low and high flow rate events.

4.27 The flow control devices (Hydrobrake or similar) utilise a pivoting by-pass door that can be activated without requiring access to the chamber, allowing the chamber and network to drain down in the event of a blockage.

4.28 Vortex flow controls generally require little maintenance as they have no moving parts, excluding the pivot door. Blockages will usually occur at the intake rather than in the device itself. The flow control chambers also feature a sump prior to the flow control to capture any sediment.

Figure 4-5: Hydro-Brake - Source: Hydro International

4.29 Inspection and maintenance to be carried out as detailed in the table below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Inspect and identify any problems with the device and the inlets / outlets.	Monthly (for the first 3 months) then every 6 months.
	Remove sediment from upstream sump	Every 6 months or as required
	Hose down unit / chamber	Every 6 months, or as required
Occasional Maintenance	Clear blockages and pivot bypass door if required to drain down chamber.	As required

Table 4-5: Flow Control Maintenance Schedule

Headwalls

4.33 Headwalls are placed at the interface between a piped network and SuDS features, most commonly basins, swales and similar. The site contains 19 headwalls.

4.34 The headwalls are pre-formed off site (Althon or similar) with additional fixtures either pre-fitted before delivery or fitted on site via suitable fixings installed on the headwall off site. On-site installation will be limited to attaching to these pre-installed fixings and no drilling, shaping, or machining of the headwalls or additional fittings should be required for installation. All fittings

4.35 In addition, any replacement of parts can be achieved by unbolting the install fixture and replacing with a new one. Minimal on-site construction or assembly required.

4.36 Rip rap is to extend in front of the apron of headwalls forming the inlets to the basins.

Figure 4-6: Precast Concrete Headwall - Source: Althon

4.37 Inspection and maintenance to be carried out as detailed in the table below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Inspect and identify any problems with the concrete headwall	Annually
	Inspect and identify any problems with the grilles, grating and flap valves. Remove any detritus.	Every 6 months or as required
	Inspect and identify any problems with the rip rap. Remove any detritus.	Annually or as required
Occasional Maintenance	Clean headwall	As required
Remedial Actions	Repair to concrete headwall for minor damage on the corners using an epoxy compound such as “Mason Mate 0868PR380 Polyester resin” or similar	As required
	Replacement of Kee Klamp, Flap valves, Grillesand grating	As required

Table 4-6: Headwall Maintenance Schedule

Emergency Events, Contact Details & Financing

4.41 In the event of a major storm, flood or otherwise unusual circumstances that could lead to damage to, or

sub-standard operation of the system, inspection of the system is to be carried out at the first possible opportunity to identify any potential remedial action that is required. This requirement will supersede the normal requirements set out in the maintenance schedules for the individual elements until such a time as the system has been confirmed to be in suitable condition and all remedial actions have been undertaken.

4.42 In addition to the inspections carried out by the adopting authorities or management company, residents should be encouraged to report any issues they may see to the relevant body.

4.43 Contact details for the relevant parties are set out below:

Name	Description	Contact number
Contractor - tbc	Adopting the piped systems and the inlet/outlet headwalls for the basins.	Homeowner Emergencies: tbc
		Business and Developers Emergencies tbc
Contractor - tbc	Management company for the basins and public open spaces	Office number: tbc
Althon	Headwall supplier	General contact 01603 488700
Hydro International	Flow Control supplier	Flow control & Flood protection contact 01275 878371

Table 4-7: Contact Details

4.44 Prior to the handover to the management company, the Contractor will be responsible for maintenance during construction.

4.45 Information and record keeping regarding maintenance activities and details of issues to be kept by the organisation responsible for maintaining the relevant element of the drainage.

5 Water Quality

5.1 Impermeable surfaces collect pollutants from a wide variety of sources including cleaning activities, wear from car tyres, vehicle oil and exhaust leaks and general atmospheric deposition. The implementation of SuDS in development drainage provides a significant benefit in removal of pollutant from development run-off.

5.2 The SuDS Manual C753 describes a 'Simple Index Approach' for assessing the pollution risk of surface run-off to the receiving environment using indices for likely pollution levels for different land uses and SUDS performance capabilities. To deliver adequate treatment SuDS components have been selected to ensure the pollution hazards are effectively treated.

5.3 The conceptual drainage proposals have been developed in a manner that will allow the site wide system to be designed to encourage passive treatment of discharged flows and to improve the water quality by removing the low-level silts, oils and metals associated with urban run-off. The design of the drainage system accords with the expectations of SuDS systems and will include a number of elements within the drainage train. Final design will provide for appropriate geometry and planting to seek maximise this benefit. Elements of the system may include any source control within development parcels, the collection and deposition of silt through the design of the highway gullies and sewer network system (as is normally and effectively the case within SuDS systems). Planting within the basins ahead of the basin inlet headwalls will also provide for a further level of water quality control.

5.4 The reserved matters approval also provides for reed beds to be introduced as part of the water quality management system in each basin. As for the detention features themselves the provision of open channel outfalls to the ordinary watercourse receptors provide a further measure to assist in managing water quality.

5.5 The SuDS storm water management features to be implemented will be designed to enhance the biodiversity and landscape character of the site, while also providing amenity space and acting as a functional feature to control storm discharges from the site and improve water quality.

5.6 Collectively the design of the overall drainage system will ensure a broadly based approach to enhancing water quality prior to discharge consistent with the FRA.

6 Next Steps and Recommendations

6.1 This report includes the detailed design drawings and calculation to support the proposed reserved matters application.

6.2 The proposed adoptable drainage system will be subject to technical approval with Swindon Borough Council and the appointed NAV under the respective S38 and S104 Agreements.

6.3 All works to existing watercourse are subject to approval with LLFA. Initial submission has been made in July 2024 to allow consents to be made.

6.4 The contractor is to manage temporary surface water management in accordance with the approved CEMP.

7 Disclaimer

7.1 The conclusions and recommendations contained herein are limited to those given the general availability of background information and the planned usage of the site.

7.2 Third party information has been used in the preparation of this report, which Brookbanks, by necessity assumes is correct at the time of writing. While all reasonable checks have been made on data sources and the accuracy of data, Brookbanks accepts no liability for same.

7.1 The benefits of this report are provided solely to Vistry and Taylor Wimpey for the proposed development *Woodlands* only.

7.2 Brookbanks excludes third party rights for the information contained in the report.

Appendix A - Approved Site Wide Drainage Strategy

Appendix B – Proposed Drainage Design Drawings

Appendix C – Flow Calculations

Appendix D- Drainage Details

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